

Suhas Girish

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Summary

Backend Software Engineer with 3 years at Mercedes-Benz building FastAPI microservices for vehicle diagnostic systems. Expertise in PostgreSQL optimization, Redis caching, Kubernetes deployments, and AI/ML integration (OpenAI API, TensorFlow Serving, BERT). Reduced API latency by 50% and database query time by 60% through performance tuning and system design. Experienced in building scalable microservices handling 10K+ connected vehicles and 2M+ monthly ML inference requests.

Skills

Programming: Python, SQL

Backend: FastAPI, REST APIs, Microservices, Async Programming, Apache Kafka, RabbitMQ

Frontend, Testing: React.js, PyTest, Unit Testing, Integration Testing, API Testing, Mocking

Databases: PostgreSQL, Redis

Infrastructure: Docker, AWS, CI/CD (GitHub Actions)

AI/ML: OpenAI API, TensorFlow Serving, BERT, ML Inference Pipelines

Work Experience

Backend Engineer — Mercedes-Benz RDI

Aug 2025 – Present

- Built FastAPI microservices for Mercedes' Vehicle Diagnostic Platform supporting 5 engineering teams across Germany and India running 200+ ECU simulations daily for predictive maintenance workflows.
- Reduced P99 latency from 800ms to 400ms through async request handling and Redis caching, improving turnaround time for ML-based fault prediction models.
- Optimized PostgreSQL queries using indexing and partitioning, reducing report generation time by 60% for real-time analytics dashboards.
- Developed internal React.js dashboards for diagnostic monitoring and API health visualization.
- Integrated OpenAI GPT-4 API for automated diagnostic summarization, reducing engineer review time by 35%.
- Developed PyTest-based unit and integration test suites for FastAPI services covering async workflows and database interactions.
- Set up Grafana dashboards and CI/CD pipelines with GitHub Actions; maintained 98.5% uptime across deployments.
- Debugged Redis cache stampede during peak load (3K+ requests); implemented Redis-based distributed locking and cache warming to stabilize traffic and reduce latency spikes by 70%.

Graduate Engineer Trainee — Mercedes-Benz RDI

Aug 2023 – Aug 2025

- Developed orchestration APIs processing 50GB+ vehicle telemetry weekly from 10K+ connected vehicles for predictive maintenance workflows.
- Built ML inference integration layer connecting FastAPI with TensorFlow Serving, handling 2M+ monthly requests with 500ms latency.
- Implemented Redis caching reducing PostgreSQL load by 40% and improving response times by 200ms.
- Profiled Python services and optimized serialization/query performance, achieving 25% throughput improvement supporting 2x traffic growth.
- Built lightweight React.js interfaces for internal diagnostic workflows and API testing utilities.
- Wrote PyTest test cases for REST APIs and failure scenarios using mocks and fixtures.
- Integrated Java legacy systems with Python microservices using REST APIs and RabbitMQ for async processing.

Software Developer Intern — Mercedes-Benz RDI

Jan 2023 – Jun 2023

- Built FastAPI data ingestion endpoints handling 10K+ diagnostic records daily with Pydantic validation for BERT-based NLP pipelines.
- Integrated fine-tuned BERT models with async inference and request batching, maintaining 200ms P95 latency for multilingual diagnostic processing.

Education

JSS Science and Technological University, Mysore

2019 – 2023

Bachelor of Engineering in Computer Science